

CONFORMAL PREDICTION FOR AUTOREGRESSIVE MODELS

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ABSTRACT

Reliable uncertainty quantification is essential in time-series prediction, particularly in high-stakes domains such as healthcare, finance, and energy systems, where predictive errors may lead to significant consequences. While conformal prediction (CP) provides finite-sample coverage under minimal assumptions, its extension to dependent data, such as autoregressive (AR) models, remains challenging. This paper introduces Conformal Prediction for Autoregressive Models (CPAM), a computationally efficient and theoretically grounded framework that integrates split conformal prediction with classical time-series methods. CPAM combines Yule-Walker estimation, autoregressive order selection guided by the Akaike Information Criterion, and residual-based calibration to construct prediction intervals with provable marginal coverage guarantees. The procedure is analytically tractable, avoids model refitting, and scales efficiently to large datasets. Empirical evaluations on both synthetic and real-world time series, including wind power and solar irradiance, demonstrate that CPAM consistently achieves empirical coverage close to the nominal target.

Index Terms— Conformal Prediction, Uncertainty Quantification, Time Series Analysis, Autoregressive Models

1. INTRODUCTION

Time-series data are central to many scientific and engineering domains, including finance, healthcare, and energy systems, as well as robotics, climate modeling, and transportation [1, 2, 3, 4, 5, 6]. These datasets exhibit temporal dependencies that enable prediction but also require careful modeling. In high-stakes applications, prediction errors can lead to serious consequences, such as resource shortages in healthcare or instability in power grids [7]. Consequently, reliable uncertainty quantification is essential for risk-aware decision-making and efficient resource allocation. While modern prediction methods, including machine learning and ensemble approaches [8, 9], have improved point prediction accuracy, estimating uncertainty with formal guarantees remains a challenge. This need is especially pressing in trustworthy AI systems, where robustness, transparency, and coverage are required [10]. These concerns have driven growing interest in conformal prediction (CP), a distribution-free framework for constructing prediction sets with finite-sample coverage under minimal assumptions, such as exchangeability [11].

CP operates as a post-hoc calibration layer that requires no changes to the underlying predictive model [12]. It has been successfully applied to classification, regression, and structured prediction, with use cases in medical diagnostics [13], drug discovery [14], computer vision [15], large language models [16], finance [17], and autonomous systems [18]. To improve scalability, variants such as split conformal [19], cross-conformal [20], and online CP [21] have been developed, enabling deployment in large-scale and nonstationary

settings. Recent advances further integrate CP with expressive prediction architectures. Conformalized quantile regression (CQR) [22] adapts split conformal inference to heteroscedastic models. CP has also been extended to graph neural networks [23, 24], transformers for sensor streams [25], and sequential portfolio optimization [26]. These efforts illustrate CP’s versatility and highlight the need for methods that retain theoretical guarantees while scaling to modern, high-dimensional, and nonstationary time-series settings.

Adapting CP to time-series data introduces challenges due to temporal dependence, which violates exchangeability. Early work addressed this issue by detecting distributional shifts using martingales [27, 28]. In prediction contexts, methods such as sliding-window calibration [29] and adaptive score rescaling [30] have been proposed. These approaches provide both theoretical and empirical support for conformal inference under temporal dependence.

Building on the foundations of time series analysis, we propose Conformal Prediction for Autoregressive Models (CPAM), a self-contained procedure that integrates split conformal prediction with classical time-series prediction techniques. CPAM begins by partitioning the observed data sequence into disjoint training and calibration datasets. It estimates autoregressive (AR) coefficients on the training segment using the Yule-Walker equations, and then transforms one-step-ahead residuals from the calibration segment into a data-driven conformal radius, ensuring finite-sample marginal coverage for the next observation. When the autoregressive order is unknown, an Akaike Information Criterion (AIC) guided extension selects the optimal order from a candidate grid up to a specified maximum order and then refits the model to the full training data. This approach preserves the closed-form efficiency of Yule-Walker estimation while maintaining formal coverage guarantees. The resulting prediction interval retains the analytical tractability of linear time-series models and remains valid under temporal dependence. Extensive empirical evaluations on both synthetic and real-world time series demonstrate that CPAM reliably achieves nominal coverage.

The remainder of this paper is organized as follows. Section 2 introduces the split conformal framework and its foundational concepts. Section 3 presents the CPAM methodology, describes its order-selection and prediction components, and establishes its theoretical coverage guarantees. Section 4 provides empirical validation using both synthetic and real-world datasets. Finally, Section 5 discusses practical implications and outlines directions for future work.

2. BACKGROUND ON SPLIT CONFORMAL PREDICTION

Let $\{(X_i, Y_i)\}_{i=1}^n \subseteq \mathcal{X} \times \mathcal{Y}$ denote a dataset of n i.i.d. observations, and fix a nominal coverage $(1 - \alpha)$, where $0 < \alpha < 1$. Split conformal prediction proceeds by partitioning the data into two disjoint subsets: a training set of size n_1 and a calibration set of size $n_2 = n - n_1$. A predictive model $\hat{f}$ is fitted using only the training set, thereby preserving statistical independence between the fitted values and the calibration responses.

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For each calibration point (X_j, Y_j) , define the absolute residual

$$\hat{\epsilon}_j = |Y_j - \hat{f}(X_j)|, \quad j = 1, \dots, n_2.$$

Let $\hat{q}_\alpha$ denote the $\lceil (1-\alpha)(1+n_2) \rceil$ empirical quantile of the residuals $\{\hat{\epsilon}_j\}_{j=1}^{n_2}$. For a new test input X_{n+1} , the split conformal prediction interval is given by

$$\mathcal{C}(X_{n+1}) = [\hat{f}(X_{n+1}) - \hat{q}_\alpha, \hat{f}(X_{n+1}) + \hat{q}_\alpha],$$

which satisfies the finite-sample marginal coverage guarantee

$$\mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1})) \geq 1 - \alpha.$$

This construction provides a computationally efficient alternative to full conformal methods, requiring only a single model fit and no refitting at test time. The choice of split ratio $n_1 : n_2$ governs a trade-off between model accuracy and calibration stability: a larger training set may improve predictive performance, while a larger calibration set yields more reliable quantile estimates. Although this procedure guarantees marginal coverage under exchangeability, its direct application to non-exchangeable settings, such as time-series data, can result in coverage violations. This limitation motivates the development of specialized adaptations, as pursued in this work.

3. CONFORMAL PREDICTION FOR TIME SERIES

Conformal prediction for time-series data must tackle two primary challenges simultaneously: (i) learning the temporal structure of the data-generating process, and (ii) constructing prediction sets that achieve valid finite-sample coverage despite temporal dependence. We consider an Autoregressive (AR) model of order p , denoted $\text{AR}(p)$, specified by

$$X_t = \phi_1 X_{t-1} + \dots + \phi_p X_{t-p} + \epsilon_t,$$

where $\Phi = (\phi_1, \dots, \phi_p)$ are the AR coefficients and ϵ_t is white noise. Given a realized trajectory $\mathcal{X}_n = (X_1, \dots, X_n)$, we denote by $\hat{X}_{n+1}$ the one-step-ahead prediction of X_{n+1} obtained from the fitted AR model. The goal is to construct a prediction interval $\mathcal{C}_\alpha^n(X_{n+1}) = [\hat{X}_{n+1} - \hat{q}_\alpha^n, \hat{X}_{n+1} + \hat{q}_\alpha^n]$ such that $\Pr(X_{n+1} \in \mathcal{C}_\alpha^n(X_{n+1})) \geq 1 - \alpha$, while making no assumptions about the noise distribution beyond the mild mixing conditions required for Yule-Walker estimation.

This section introduces *Conformal Prediction for Autoregressive Models* (CPAM), a theoretically grounded yet computationally efficient procedure that integrates classical time-series modeling with split conformal prediction. The approach is notable for combining closed-form AR estimation, data-driven order selection, and residual-based conformal calibration in a single, modular pipeline. CPAM requires only two linear-algebraic operations beyond standard AR fitting and scales efficiently to large datasets. It provides predictive intervals with finite-sample marginal coverage guarantees, even when the true AR order is unknown and must be estimated from data. To clarify the computational flow, we present CPAM as a composition of three stages: (1) coefficient estimation via the Yule-Walker equations, (2) autoregressive order selection guided by the Akaike Information Criterion (AIC), and (3) residual calibration via the split conformal prediction. This design enables transparent analysis, easy implementation, and broad applicability in real-world time-series prediction tasks.

3.1. Coefficient Estimation via Yule-Walker

The Yule-Walker equations provide a closed-form solution for estimating the coefficients and innovation variance of an AR model. This method relies solely on empirical autocovariances and is particularly effective for stationary time series with moderate order [1]. Within the CPAM framework, it forms a core component of both the order selection and prediction interval construction procedures, as detailed in Algorithm 1. Specifically, Line 2 computes the sample autocovariances $\hat{\rho}_k$ for lags $k = 0, \dots, p$. These estimates define the Toeplitz autocovariance matrix¹ $\mathbf{R}$ (Line 4) and the corresponding right-hand side vector $\mathbf{r}$ (Line 5) in the Yule-Walker system. Solving the linear system $\mathbf{R}\hat{\Phi} = \mathbf{r}$ yields the estimated coefficient vector $\hat{\Phi}$ (Line 6). The innovation variance $\hat{\sigma}^2$ is then computed by subtracting the variance explained by the fitted model from the total variance (Line 7).

Algorithm 1 Yule-Walker($\mathcal{X}_n, p$)

Require: Samples $\mathcal{X}_n = (X_1, \dots, X_n)$; AR order p .

Ensure: Coefficient vector $\hat{\Phi}$; innovation variance $\hat{\sigma}^2$

- 1: **for** $k \leftarrow 0$ **to** p **do**
 - 2: $\hat{\rho}_k \leftarrow \frac{1}{n} \sum_{t=k+1}^n X_t X_{t-k}$
 - 3: **end for**
 - 4: $\mathbf{R} \leftarrow \text{Toeplitz}([\hat{\rho}_0, \dots, \hat{\rho}_{p-1}])$
 - 5: $\mathbf{r} \leftarrow [\hat{\rho}_1, \dots, \hat{\rho}_p]^\top$
 - 6: $\hat{\Phi} \leftarrow \mathbf{R}^{-1} \mathbf{r}$
 - 7: $\hat{\sigma}^2 \leftarrow \hat{\rho}_0 - \hat{\Phi}^\top \mathbf{r}$
 - 8: **return** $(\hat{\Phi}, \hat{\sigma}^2)$
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3.2. Order Selection via Akaike Information Criterion

In many practical settings, the AR order p is unknown and must be selected from the data. To this end, we adopt a procedure based on the Akaike Information Criterion (AIC) applied to Yule-Walker estimates, referred to as AIC-YW. This method evaluates a grid of candidate orders $p \in \{0, \dots, p_{\max}\}$ and selects the value minimizing the AIC score

$$\text{AIC}(p) = 2(p+1) + (n-p) \log(\hat{\sigma}_p^2),$$

where $\hat{\sigma}_p^2$ is the innovation variance estimated via the Yule-Walker equations at order p [31]. The procedure is summarized in Algorithm 2. For each candidate order p , Line 2 fits the $\text{AR}(p)$ model using the Yule-Walker estimator and computes the corresponding innovation variance. Line 3 evaluates the AIC score at that order. After scanning all values in the search range, Line 5 selects the order with the minimal AIC. Finally, Line 6 refits the selected model on the entire dataset and returns the estimated coefficients $\hat{\Phi}$ along with the selected order $\hat{p}$.

3.3. Conformal Prediction for Autoregressive Models (CPAM)

We now integrate the preceding components into a complete prediction procedure, Conformal Prediction for Autoregressive Models (CPAM), designed to produce valid prediction intervals under weak temporal dependence. CPAM employs split conformal inference and consists of three primary stages: model training, residual calibration, and interval construction. The procedure is detailed in Algorithm 3. The input sequence $\mathcal{X}_n$ is divided into a training dataset of size n_{tr}

¹A Toeplitz matrix $\mathbf{R}$ is a structured matrix whose entries satisfy $R_{i,j} = \hat{\rho}_{|i-j|}$ for $1 \leq i, j \leq p$, i.e., each element depends only on the lag $|i-j|$, so all diagonals parallel to the main diagonal are constant.

Algorithm 2 AIC-YW($\mathcal{X}_n, p_{\max}$)**Require:** Samples $\mathcal{X}_n = (X_1, \dots, X_n)$, maximal lag $p_{\max}$.**Ensure:** AR order $\hat{p}$, AR coefficients $\hat{\Phi}$.

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1: for  $p \leftarrow 0$  to  $p_{\max}$  do
2:    $(\hat{\Phi}^{(p)}, \hat{\sigma}_p^2) \leftarrow \text{Yule-Walker}(\mathcal{X}_n, p)$ 
3:    $\text{AIC}(p) \leftarrow 2(p+1) + (n-p) \log(\hat{\sigma}_p^2)$ 
4: end for
5:  $\hat{p} \leftarrow \arg \min_{0 \leq p \leq p_{\max}} \text{AIC}(p)$ 
6:  $(\hat{\Phi}, \hat{\sigma}^2) \leftarrow \text{Yule-Walker}(\mathcal{X}_n, \hat{p})$ 
7: return  $(\hat{p}, \hat{\Phi})$ 

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and a calibration dataset of size $n_{\text{cal}} = n - n_{\text{tr}}$ (Line 1). If the AR order p is known, the coefficients are estimated using the Yule-Walker method (Lines 2-4); otherwise, the order is selected via AIC-YW (Line 6). One-step-ahead residuals on the calibration set are computed from the fitted AR model and used as conformity scores (Lines 8-9). The prediction for X_{n+1} is formed using the most recent p values and the estimated coefficients (Line 11). The prediction interval is constructed by computing the upper empirical α -quantile of the residual magnitudes (Line 12) and centering a symmetric interval around the point prediction (Line 13). The resulting interval C_α^n achieves marginal coverage at least $1 - \alpha$.

Algorithm 3 Conformal Prediction for Autoregressive Models (CPAM)**Require:** Sample sequence $\mathcal{X}_n = \{X_1, \dots, X_n\}$; training size n_{tr} ; AR order p (or maximal order $p_{\max}$); coverage $(1 - \alpha)$; a new test point X_{n+1} **Ensure:** Prediction Interval C_α^n for X_{n+1}

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1:  $\mathcal{X}_{n_{\text{tr}}} \leftarrow \{X_1, \dots, X_{n_{\text{tr}}}\}$ ,  $\mathcal{X}_{n_{\text{cal}}} \leftarrow \{X_{n_{\text{tr}}+1}, \dots, X_n\}$ 
2: if order  $p$  is given then
3:   Set  $\hat{p} \leftarrow p$ 
4:    $\hat{\Phi}^{(\text{tr})} \leftarrow \text{Yule-Walker}(\mathcal{X}_{n_{\text{tr}}}, \hat{p})$ 
5: else
6:    $(\hat{p}, \hat{\Phi}^{(\text{tr})}) \leftarrow \text{AIC-YW}(\mathcal{X}_{n_{\text{tr}}}, p_{\max})$ 
7: end if
8: for  $t \leftarrow n_{\text{tr}} + \hat{p} + 1$  to  $n$  do
9:    $\hat{\epsilon}_t \leftarrow X_t - \phi_1^{(\text{tr})} X_{t-1} - \dots - \phi_{\hat{p}}^{(\text{tr})} X_{t-\hat{p}}$ 
10: end for
11:  $\hat{X}_{n+1} \leftarrow \phi_1^{(\text{tr})} X_n + \dots + \phi_{\hat{p}}^{(\text{tr})} X_{n-\hat{p}+1}$ 
12:  $\hat{q}_\alpha^n \leftarrow$  the upper  $\alpha$ -quantile of  $\{|\hat{\epsilon}_{n_{\text{tr}}+\hat{p}+1}|, \dots, |\hat{\epsilon}_n|\}$ 
13:  $C_\alpha^n \leftarrow [\hat{X}_{n+1} - \hat{q}_\alpha^n, \hat{X}_{n+1} + \hat{q}_\alpha^n]$ 
14: return  $C_\alpha^n$ 

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4. EXPERIMENTAL RESULTS

We evaluate CPAM on both real-world and synthetic time series. The source code for the implementation of our algorithm is available at <https://github.com/byaghooti/CPAM>.

4.1. Synthetic Datasets

All experiments use a nominal coverage of $1 - \alpha = 0.95$ and adopt CPAM, as described in Section 3. For each study, we report empirical coverage, defined as the proportion of the points that fall inside the prediction interval. To assess finite-sample performance under controlled conditions, we simulate AR(p) processes of varying sample size and innovation distribution. Let $\Phi = (\phi_1, \phi_2, \phi_3, \phi_4) =$

$(0.4, 0.2, 0.1, -0.2)$, and define AR(p) to use the first p components of ϕ . Innovations ϵ_t are drawn i.i.d. from one of three distributions: $\mathcal{N}(0, 1)$, centered Exp(1), or centered and variance-normalized Lognormal(0, 1). For each combination of sample size $n \in \{60, 100, 200, 500\}$, order $p \in \{1, 2, 3, 4\}$, and innovation law, we generate 10,000 Monte-Carlo replicates.

Table 1 reports coverage when the data is divided evenly between training and calibration ($n_{\text{tr}} : n_{\text{cal}} = 1 : 1$). Across all settings the empirical coverage stays within one percentage point of the nominal 95%, confirming the finite-sample guarantee even for heavy-tailed log-normal noise. Table 2 repeats the experiment with a 1 : 3 train-calibration ratio.

IID $\mathcal{N}(0, 1)$ innovations				
n	AR(1)	AR(2)	AR(3)	AR(4)
60	96.72	96.52	96.25	96.37
100	96.14	95.45	95.76	95.88
200	95.59	95.74	95.72	95.67
500	94.74	94.86	94.88	94.90

IID Exp(1) innovations (centered)				
n	AR(1)	AR(2)	AR(3)	AR(4)
60	96.26	96.30	95.90	96.27
100	95.71	95.77	95.94	95.48
200	94.78	95.19	95.41	95.72
500	94.54	95.01	95.03	95.11

IID Lognormal(0, 1) innovations (centered)				
n	AR(1)	AR(2)	AR(3)	AR(4)
60	96.28	96.04	95.83	96.44
100	95.70	95.45	95.37	95.82
200	95.01	95.59	95.54	95.74
500	94.99	94.69	95.16	95.12

Table 1: Empirical coverage (%) with a 1 : 1 train-calibration split.

IID $\mathcal{N}(0, 1)$ innovations				
n	AR(1)	AR(2)	AR(3)	AR(4)
60	95.72	95.60	95.20	95.54
100	96.14	95.65	95.79	96.00
200	95.74	95.27	95.35	95.14
500	94.93	94.94	95.05	94.96

IID Exp(1) innovations (centered)				
n	AR(1)	AR(2)	AR(3)	AR(4)
60	95.51	95.20	95.02	95.51
100	95.75	95.98	95.95	95.72
200	95.11	95.00	94.94	95.27
500	94.72	95.08	95.10	95.24

IID Lognormal(0, 1) innovations (centered)				
n	AR(1)	AR(2)	AR(3)	AR(4)
60	95.47	94.81	94.87	95.43
100	95.86	95.33	95.83	96.02
200	95.15	95.44	95.28	95.21
500	95.02	94.71	95.28	95.03

Table 2: Empirical coverage (%) with a 1 : 3 train-calibration split.

Across all the simulated time points, CPAM consistently delivers coverage at or above the nominal 95% level. The slight under-coverage observed for very small n and highly skewed noise disappears once $n \geq 100$, suggesting that the weak-mixing approximation is adequate in typical moderate-sample regimes.

Figure 1 shows CPAM’s one-step-ahead predictions for two synthetic AR processes: an AR(1) model with coefficient $\phi_1 = 0.8$ (left panel) and an AR(2) model with coefficients $(\phi_1, \phi_2) = (0.5, -0.3)$ (right panel). For each series, the AR order is selected by AIC on a training segment, parameters are estimated via the Yule-Walker equations, and prediction intervals are calibrated on a hold-out set using the $(1 - \alpha)$ quantile of absolute residuals (here $\alpha = 0.05$). The blue curve denotes the ground-truth values, the orange curve the CPAM predicted values, and the shaded area the resulting prediction interval. The method closely tracks the underlying dynamics in both cases while maintaining the nominal coverage, with modestly wider bands for the AR(2) process reflecting its higher effective complexity.

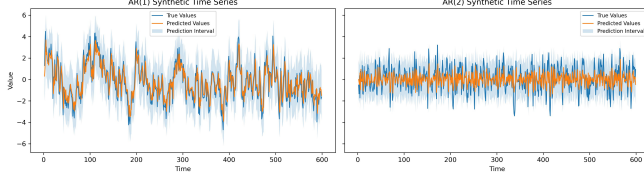


Fig. 1: CPAM one-step-ahead prediction with 95% coverage for synthetic AR(1) (left) and AR(2) (right) models.

4.2. Real-World Applications

In this subsection, we evaluate CPAM on two real-world renewable energy datasets under a nominal coverage of 99%.

Figure 2 illustrates CPAM applied to the Hackberry wind power dataset [32], using hourly generation data from 12:00 to 17:00. The model is trained on data from May to December 2019 and tested on the corresponding period in 2020. The blue curve denotes the observed values, the orange curve the predicted values, and the shaded area the resulting prediction interval. Each subplot reports the empirical coverage in its title, which ranges from 97.8% to 98.9%. Prediction interval widths adapt to hours with more volatile wind conditions, typically in the early afternoon, while remaining sufficiently tight during more stable periods.

Figure 3 presents one-hour-ahead predictions for Direct Normal Irradiance (DNI) in Mountain View. CPAM is trained on the first six months of 2018 (January-June) and tested on the subsequent three months (July-September). Each subplot displays the observed DNI (blue line), the point prediction (orange line), and the prediction interval (shaded area). The subplot titles report the empirical coverage, i.e., the fraction of test points contained within the interval, which ranges from 98.9% to 100.0%, except for one experiment (96.4%) caused by sudden fluctuations in DNI over short intervals. These results closely match the nominal 99% level. We also observe that intervals are slightly wider between 15:00 and 16:00, reflecting increased irradiance variability around peak solar insolation.

Across both datasets, CPAM consistently achieves near-nominal coverage without requiring bootstrapping or model refitting, relying instead on a single Yule-Walker solve and one quantile computation. These results confirm its finite-sample validity and computational efficiency for uncertainty quantification in renewable energy prediction.

5. CONCLUSION

This paper has introduced CPAM, a conformal prediction framework specifically designed for autoregressive models. The method integrates classical linear estimation with split conformal calibration to produce prediction intervals that retain validity under weak dependence. By employing the Yule-Walker equations for coefficient

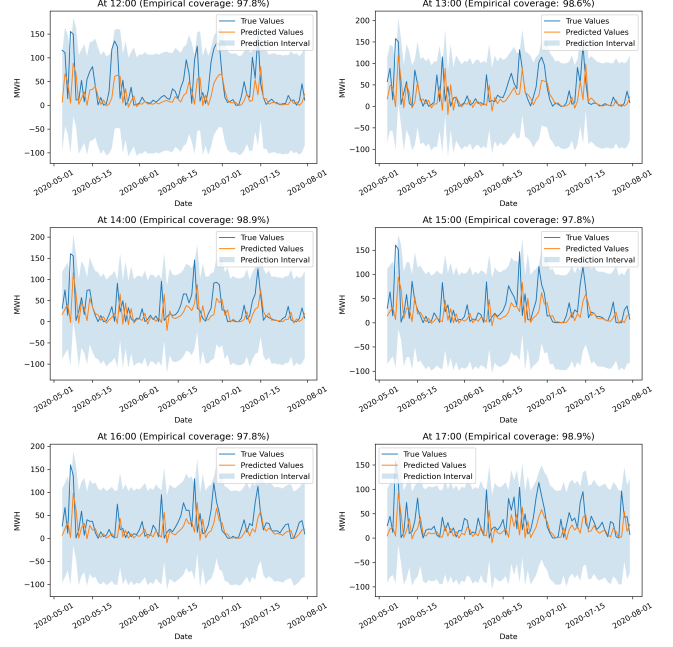


Fig. 2: CPAM’s performance on Wind Power Generation (Hackberry dataset).

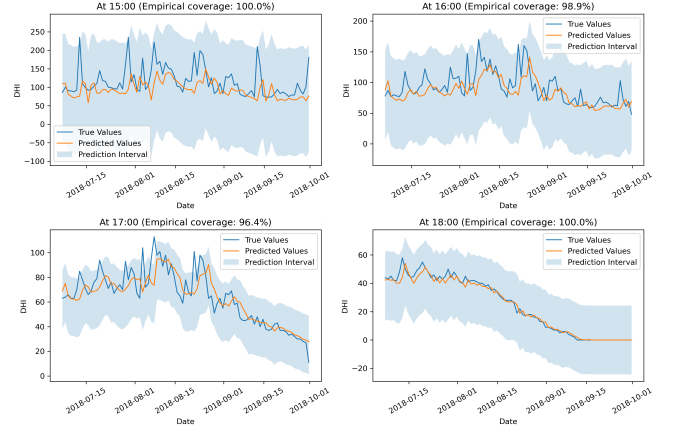


Fig. 3: CPAM’s performance on Solar Energy (Direct Normal Irradiance) in Mountain View.

estimation and autoregressive order selection guided by the Akaike Information Criterion, CPAM achieves analytical tractability and computational efficiency without the need for model retraining or resampling. Theoretical analysis establishes finite-sample marginal coverage guarantees, while empirical evaluations on synthetic and real-world datasets, including solar irradiance and wind power generation, demonstrate coverage levels close to the nominal target across a variety of noise distributions and temporal structures. These results underscore CPAM’s effectiveness and reliability for uncertainty quantification in time-series prediction. Future research will focus on extending CPAM to multivariate settings, addressing nonstationarity, and integrating it with nonlinear prediction architectures while maintaining rigorous coverage guarantees.

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